

Ravani Roshan

AI Engineer · Full-Stack Developer · OSS Builder

Ahmedabad, Gujarat, India +91 91060 95375 ravaniroshansingh@gmail.com github.com/RavaniRoshan

SUMMARY

Computer Science Engineering undergraduate specialising in agentic AI, LLM systems, and production-grade full-stack infrastructure. Author of AgentTrace — an open-source agent observability tool published on PyPI and npm, used by developers building LangChain, CrewAI, and custom agent loops. Hands-on experience designing end-to-end systems across reasoning agents, RAG pipelines, real-time APIs, and browser automation. Strong proficiency in Python, JavaScript/TypeScript, and Rust, with emphasis on correctness, system reliability, and interpretability. Actively pursuing AI engineering internships and research engineering roles.

EDUCATION

B.Tech, Computer Science Engineering Aug 2023 – Present

Silver Oak University, Ahmedabad, Gujarat · CGPA: 8.0

EXPERIENCE

Core Developer · Google Developers Group, Ahmedabad Oct 2023 – Dec 2025

- Led engineering sessions on AI tooling and agentic system design for a community of 200+ developers; mentored peers on LLM integration patterns and production deployment practices.
- Contributed to multi-domain software projects spanning web infrastructure and applied ML, with sustained focus on correctness, maintainability, and system-level thinking.
- Maintained active engagement with emerging frameworks including LangChain, LangGraph, and Claude API tooling across community-driven build sprints.

PROJECTS

AgentTrace — *Open-Source Agent Observability & Debugging Tool* Jan 2025 – Present

- **Published OSS tool.** Designed and shipped AgentTrace — a local-first visual debugger for AI agent loops, published as a dual-package system:
- `pip install agenttrace` (PyPI) + `npx @ravaniroshan/agenttrace` (npm) — production-ready, installs in seconds, zero config required.
- Implemented three decorator-based instrumentation primitives (`@trace`, `@trace_llm`, `@trace_tool`) that capture full prompt history, token counts, per-step cost, latency, and error tracebacks without modifying agent logic.
- Built `patch_langchain()` auto-instrumentation API enabling zero-decorator tracing across LangChain, CrewAI, LlamaIndex, AutoGen, and Smolagents — supporting both sync and async agent loops via thread-safe TraceCollector.
- Architected complete dual-runtime system: Python library (FastAPI + stdlib, no heavy deps) + Node.js CLI (Express) sharing a local `~/agenttrace/traces/` JSON store — enabling cross-runtime trace inspection with a single `npx` command.
- Built Astro + Starlight documentation site at `agenttrace.dev`; published comparison analysis against LangSmith and Helicone positioning AgentTrace as the only 100% local, framework-agnostic agent debugger.

AXIOM-ONE — *Research-Grade Reasoning System* Dec 2025 – Present

- Designing a reasoning-first research agent optimised for epistemic correctness over conversational fluency — implements structured problem decomposition, multi-step iterative reasoning, and self-verification with explicit intermediate artifact exposure.
- Targets high-stakes research and analytical settings where auditability, failure detection, and reliability are critical system constraints.

LegacyLens — *AI Archaeologist for Legacy Codebases* — *Kaggle Competition* Dec 2025

- Transforms 1M+ lines of legacy source code into dynamic dependency graphs and logical execution maps using Gemini 2.0 Pro to trace control and data flow, identify structural fragility, and enable conversational interrogation of monolithic systems.

ZERO-COMP Solar Weather API — *Enterprise-Grade Solar Flare Prediction Platform* Early 2025

- Architected real-time solar flare prediction platform powered by NASA-IBM's Surya-1.0 transformer model; exposes predictive intelligence through RESTful APIs, WebSocket alerts, and a Next.js analytical dashboard with live visualisations.
- Engineered using FastAPI, Supabase (PostgreSQL + Auth), Hugging Face Transformers; deployed across Vercel and Railway/Fly.io with Docker for reproducible production environments.

Agent-X — *Browser-Native AI Automation Agent* Oct 2024 – Dec 2024

- Built a browser-native AI agent executing complex web tasks from natural-language instructions using vision–language models (Gemini 2.5); integrates Python, FastAPI, Playwright, and React with an explicit emphasis on safety, transparency, and open-source extensibility.

RAG Course Project — *Production-Grade RAG Curriculum* Jun 2024 – Sep 2024

- Authored a 15-module advanced RAG curriculum covering chunking strategies, embedding models, reranking, hybrid retrieval, vector databases, query optimisation, agentic pipelines, and deployment — targeting AI/ML practitioners seeking system-level RAG mastery.

SKILLS

Programming & Systems

Python ML pipelines, agent orchestration, FastAPI, model integration, evaluation loops.

JS / TS Full-stack integration — React, Next.js, Node.js; tight AI backend coupling.

Rust Correctness-driven systems design; safety, explicit error handling, reliable pipelines.

C++ Algorithmic intuition, performance-aware programming, memory reasoning.

AI / ML Systems

LLMs Designing, integrating, and evaluating LLM-driven systems for reasoning, automation, and research.

Agents Single- and multi-agent systems using LangGraph and custom orchestration; tool use, control flow, observability.

RAG Full RAG stacks: chunking, embedding selection, reranking, hybrid retrieval, evaluation, deployment.

Vision Prompt engineering for vision–language models; CV perception in agentic systems.

Backend, Cloud & Infrastructure

Infrastructure FastAPI, PostgreSQL, MySQL, Docker, AWS, GCP, Vercel, Railway, Fly.io, Nginx, Supabase.

Product React / Next.js dashboards, real-time WebSocket systems, product-oriented engineering.

CERTIFICATIONS

AI Systems & Agents (2024 – 2026)

- **Claude Code in Action** — Anthropic · Mar 2026
- **AI Fluency Framework & Foundations** — Anthropic · Mar 2026
- **AI Fluency for Students** — Anthropic · Mar 2026
- **Claude 101** — Anthropic · Mar 2026
- **Transformer Models and BERT Model** — Google · Mar 2026
- **Responsible AI for Developers: Privacy & Safety** — Google · Jan 2026
- **MLOps for Generative AI** — Google · Dec 2025
- **Building toward Computer Use with Anthropic** — DeepLearning.AI · Mar 2025
- **AI Agents in LangGraph** — DeepLearning.AI · 2024
- **Building AI Browser Agents** — DeepLearning.AI · 2024
- **Finetuning Large Language Models** — DeepLearning.AI · 2024
- **LangChain for LLM Application Development** — DeepLearning.AI · 2024
- **Agent Communication Protocol (ACP)** — DeepLearning.AI · 2024
- **LLMs as Operating Systems: Agent Memory** — DeepLearning.AI · 2024
- **Prompt Engineering for Vision Models** — DeepLearning.AI · Dec 2024
- **Introduction to Generative AI** — Google · Nov 2024

ML / Data Science (2024)

- **Machine Learning with Python** — Cognitive Class + freeCodeCamp · Dec 2024
- **Computer Vision** — Kaggle · Nov 2024
- **Intro to Game AI and Reinforcement Learning** — Kaggle · Nov 2024
- **Statistics 101** — Cognitive Class · Dec 2024

Software Engineering (2024 – 2025)

- **Software Engineering Job Simulations** — Forage — Electronic Arts, Walmart, Goldman Sachs · 2024
- **Front End Development Libraries** — freeCodeCamp · Feb 2025
- **Foundational C# with Microsoft** — freeCodeCamp · Nov 2024